

Xiangling Xu (许湘灵)

My Website

xu[dot]xiangling[at]inria[dot]fr

Google Scholar

Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

- 2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*
- 2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00
- 2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Gravity with Noncommutative Geometry, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

1. **X. Xu**, I. Klep, and M.-O. Renou, *Quantum tree networks: Complete semidefinite hierarchies and the source-transfer model*, 2026. arXiv: 2609.19310 [quant-ph]. [URL: https://arxiv.org/abs/2609.19310](https://arxiv.org/abs/2609.19310).
2. F. M. Kalarde, S. Boreiri, **X. Xu**, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou, *Fermions are fundamentally more nonlocal than bosons*, Jun. 11, 2026. arXiv: 2606.12363 [quant-ph]. [URL: https://arxiv.org/abs/2606.12363](https://arxiv.org/abs/2606.12363).
3. **X. Xu**, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou, *The bulk spectral gap is semi-decidable: A convergent family of certified upper bounds*, Jun. 3, 2026. arXiv: 2606.03836 [quant-ph]. [URL: https://arxiv.org/abs/2606.03836](https://arxiv.org/abs/2606.03836).
4. F. M. Kalarde, **X. Xu**, and M.-O. Renou, *Comment on "quantum theory based on real numbers cannot be experimentally falsified": On the compatibility of physical principles with information theory for fermions*, Apr. 2026. arXiv: 2604.07425 [quant-ph]. [URL: https://arxiv.org/abs/2604.07425](https://arxiv.org/abs/2604.07425).
5. M.-O. Renou, **X. Xu**, and L. T. Ligthart, "Two convergent npa-like hierarchies for the quantum bilocal scenario," *Journal of Mathematical Physics*, vol. 67, no. 1, p. 012 203, Jan. 2026, ISSN: 0022-2488. [DOI: 10.1063/5.0211008](https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf). eprint: https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf.
6. M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. [URL: https://arxiv.org/abs/2509.25145](https://arxiv.org/abs/2509.25145).
7. I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. [URL: https://arxiv.org/abs/2507.17006](https://arxiv.org/abs/2507.17006).

- 8 **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph].  URL: <https://arxiv.org/abs/2505.22309>.
- 9 X. Luo, **X. Xu**, and X. Wang, "On the kinematic morphology around haloes," *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023.  DOI: <https://doi.org/10.1093/mnras/stac3500>.
- 10 **X. Xu**, "Quantum nonlocality in bilocal networks: An operator algebraic perspective," M.S. thesis, ETH Zurich, 2023.  DOI: <https://doi.org/10.3929/ethz-b-000613670>.

Talks & Conference Activities

Invited Talks

- Jun 2026  2026 SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom
- Dec 2025  Quantum Meets, online at <https://quantum-meets.github.io>
- Jul 2025  IWOTA 2025 Twente, Twente, Netherland
- Oct 2023  Saclay Quantum Seminar, Inria Saclay, France
- Jun 2025  IQC-PCQT-Quantum Saclay Workshop, Paris, France

Contributed Talks

- Aug 2026  QCCrypt 2026, Ottawa, Canada
- Jan 2026  QIP 2026 Riga – 2 contributed talks, Riga, Latvia
- Nov 2025  3rd Colloquium GDR TeQ, Grenoble, France
- Oct 2025  YQIS25, Barcelona, Spain
- Sep 2024  Causalworlds 2024, Waterloo, Canada
- Nov 2023  Workshop Defi EQUIP, Lyon, France

Talks during group visits

- Sep 2026  Curiosity, Inria & Télécom Paris & IP Paris
- Aug 2026  Ranyiliu Chen, Quantum Science Center of Huangdong-HongKong-Macao Greater Bay Area
- Jul 2026  Penghui Yao, Nanjing University
- Feb 2026  Jean-Daniel Bancal, CEA, Institut de physique théorique
- Dec 2025  Jie Wang, AMSS-CAS, Beijing
- Nov 2025  Thomas Vidick, EPFL
- Oct 2025  Jens Eisert, Freie Universität Berlin
- Feb 2025  QAT Team, ENS Paris
- Jul 2024  William Slofstra, University of Waterloo
- Dec 2023  David Gross, University of Cologne

Poster Sessions

- Aug 2026  CQCC & GQSF 2026, Shenzhen, China
- Nov 2024  2nd Colloquium GDR TeQ, Paris, France
-  YQIS24, Paris, France

Talks & Conference Activities (continued)

Other Participation

- Sep 2026  *Quantum Information Aussois*, Aussois, France
- Mar 2025  *828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory*, Bad Honnef, Germany
- Sep 2024  *VQF Vienna Quantum Foundations Conference*, Vienna, Austria
- Nov 2023  *POP23 Workshop on Future Trends in Polynomial OPTimization*, Toulouse, France

Collaborations

Invited Group Visits

- Oct-Nov 2026  *David Gross*, University of Cologne
Invited three-week visit
- Jul 2026  *Penghui Yao*, Nanjing University
Invited one-week visit
- Dec 2025 & May 2026  *Jie Wang*, AMSS-CAS, Beijing
Two invited one-week visits
- Oct-Nov 2025  *Thomas Vidick*, EPFL
Invited two-week visit
- Sep 2024  *Igor Klep*, University of Ljubljana
Invited two-week visit funded by Erasmus+ Mobility program
- Jul 2024  *William Slofstra*, Institute for Quantum Computing, University of Waterloo
Invited two-week visit
- Dec 2023  *David Gross*, University of Cologne
Invited one-week visit

Organizations

- Mar 2025 - Mar 2026  Coorganizer for *Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory*
Coordinating the Gaspard Monge Visiting Professor program for Igor Klep
Co-planning the seminar and the schedule for invited speakers
- Mar 2025  Coorganizer for *828. WE-Heraeus-Seminar at Bad Honnef*
Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts
Coordinating with the organizers and inviting some of the speakers
- Feb 2024 - Now  Coorganizer of *Project COMPUTE for Quanteria*
Coordinating with the members, project poster and website
- Sep 2023  Coorganizer for *Saclay Quantum Seminar on NPA hierarchy*
Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

- 2026  *Erasmus+ Mobility Program*, for visiting the University of Cologne
- 2024  *Erasmus+ Mobility Program*, for visiting the University of Ljubljana
- 2020  *McNab Undergraduate Scholarship*, University of Toronto

Awards and Scholarships (continued)

2017 - 2020	■	<i>Dean's List Scholar</i> , University of Toronto
2018	■	<i>The Class of 3To and Associates Scholarship in Mathematics and Physics</i> , University of Toronto
	■	<i>International Exchange Award</i> , University of Toronto
	■	<i>Bronze Medal</i> , 2017 University Physics Competition
2017	■	<i>Eileen Doman Award</i> , University of Toronto
	■	<i>Summer Undergraduate Research Program</i> , Canadian Institute for Theoretical Astrophysics
2016	■	<i>President's Scholar of Excellence Program</i> , University of Toronto

Teaching Experience

Lecture

Dec 2025	■	<i>Recent trends in parallel, distributed, and quantum computing</i> QDCS Master, Université Paris-Saclay
Nov 2024	■	<i>Recent trends in parallel, distributed, and quantum computing</i> QDCS Master, Université Paris-Saclay

Teaching Assistance

2022-2023	■	<i>Linear Algebra I & II</i> Department of Mathematics, ETH Zürich
2021-2022	■	<i>Analysis I & II</i> Department of Mathematics, ETH Zürich

Skills

Languages	■	Chinese (native), English (fluent), German (beginner)
Coding	■	Python (Ncpol2sdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, $\text{\LaTeX}$

Referees

Available on Request